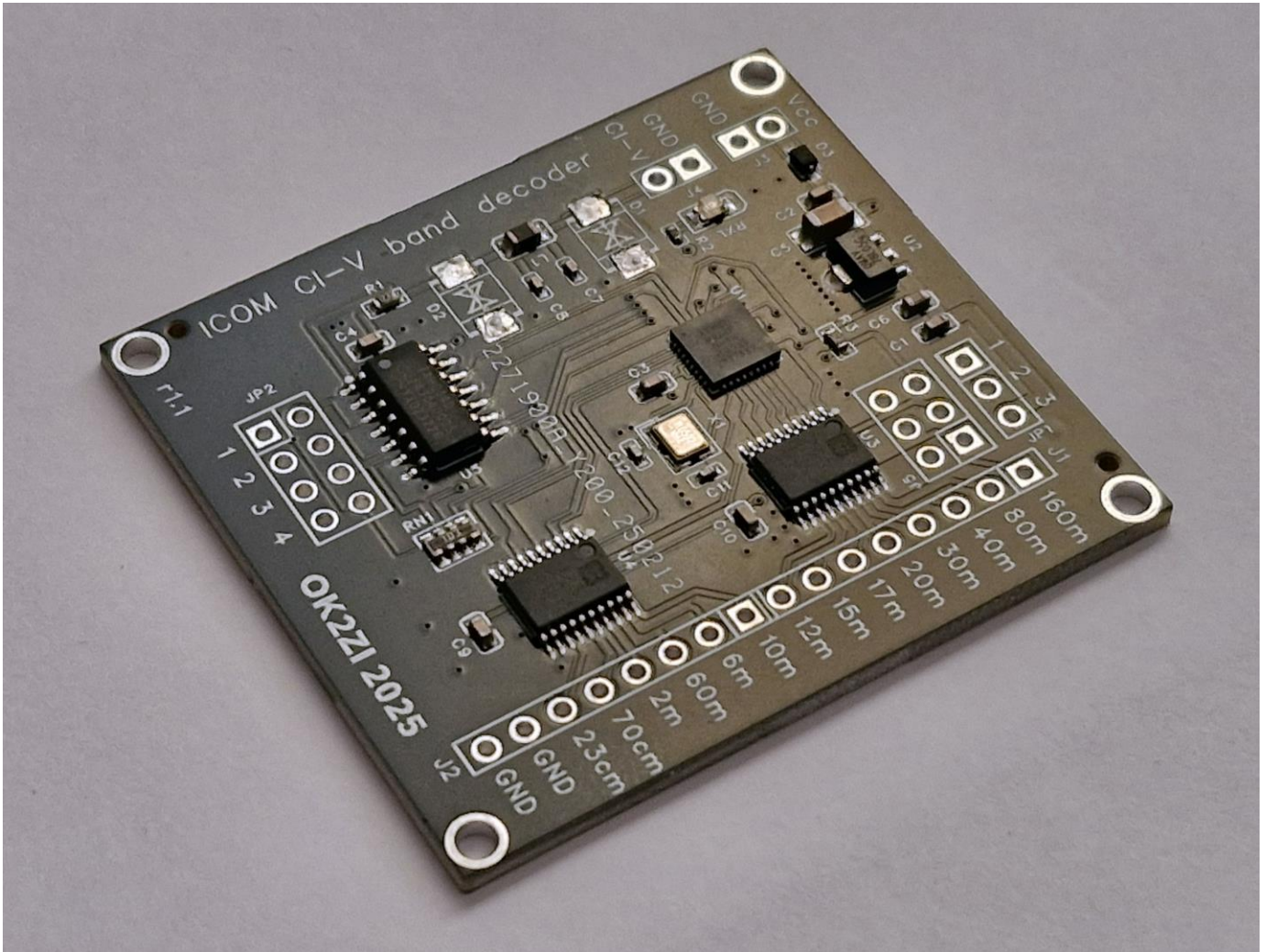


Icom CI-V band decoder with BCD output capability

April 2025

OK2ZI



Rev 1.0 (04/2025)

Introduction

The ICOM CI-V band/BCD decoder continuously decodes your transceiver's frequency changes and provides a separate output for each of the fourteen bands of your rig. The decoder passively listens ("sniffs") the CI-V bus and converts information about the current frequency to a defined single output or encodes the band info to the 4bits BCD code.

By connecting an external antenna switch to the decoder's output, automatic and correct antenna selection can be accomplished.

The ICOM CI-V decoder is primary designed for ICOM radios, but it can work with any device using the CI-V bus and the CI-V protocol defined by ICOM JAPAN company.

The ICOM CI-V decoder is purely a passive listener on the C-IV bus. This means that for the decoder to be able to decode some information about the set frequency, there must be a device on the CI-V bus that periodically queries your RIG for the frequency. This can be, for example, a station log, power amplifier, Microham interface, etc. If you do not use any such device, most ICOM transceivers can be switched to CI-V transceive mode, where the transceiver itself automatically transmits the actual frequency on the CI-V bus.

The decoder is listening for packets at CI-V bus and decodes them. When a valid packet with the current frequency is received, the appropriate output is activated and the RXL led will blink.

The ICOM CI-V decoder provides output band information using 14 positive voltage outputs. The core of this decoder is a powerful RISC microcontroller with specific software.

This software can be **customized on order**, for example another band segments, antenna tuners control, 80m YAGI segments switching etc. If you need customer specific functions, do not hesitate to contact me at email below.

Decoder board description

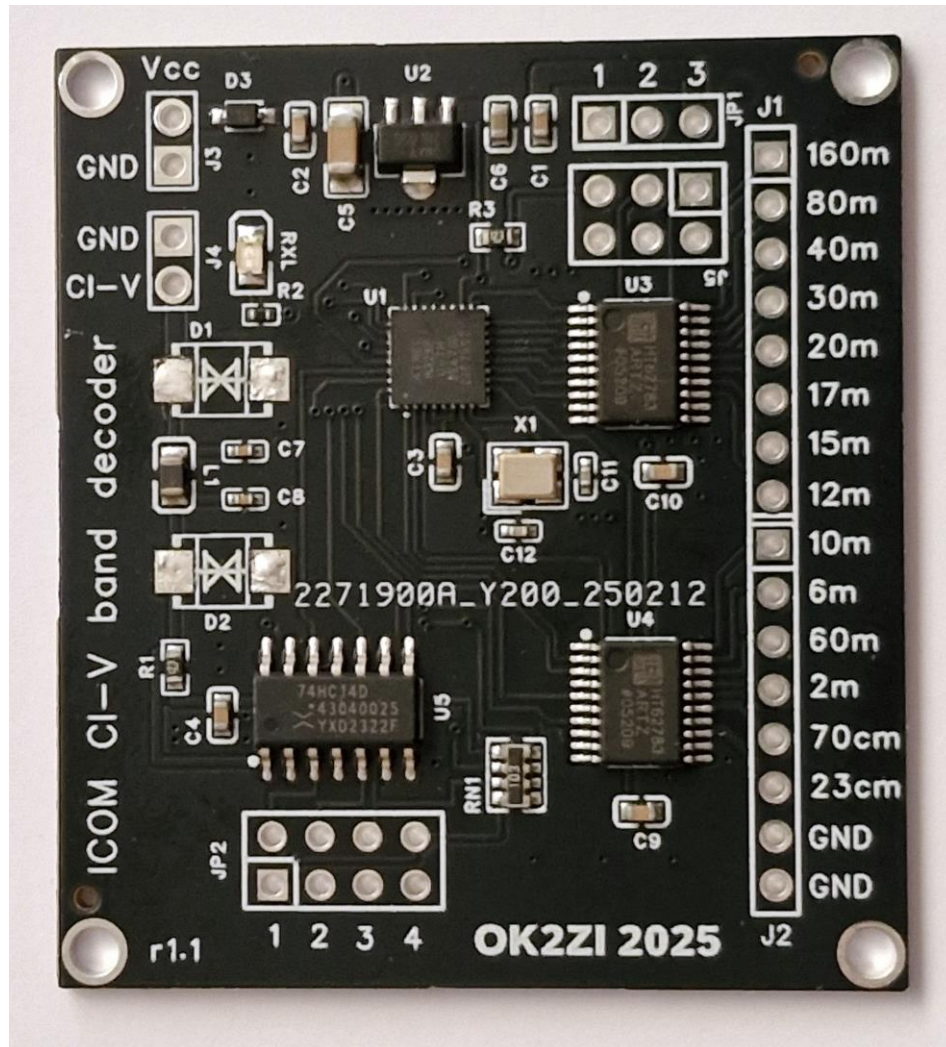


Figure 1.

- J1+J2 – outputs + GND connection
- J3 – power supply connection 13.8 - 30V max
- J4 – C-IV bus connection
- J5 – service connector
- JP1 – output drivers power supply selector
- JP2 – configuration selector

Output drivers

All 14 outputs are driven by N-channel Darlington pair transistor switches U3 and U4 HT62783ARTZ.

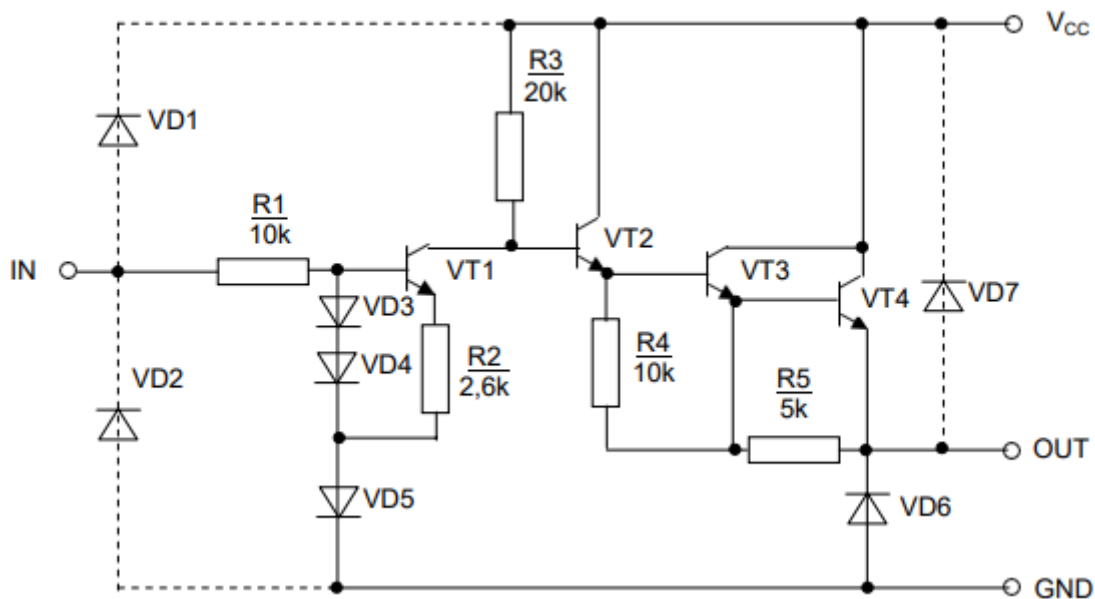


Figure 2.

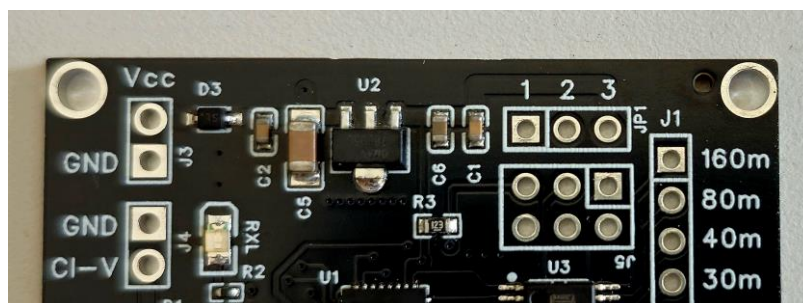
All 14 switches function in the same manner. When their input goes high, their output switches high. All unselected outputs float at ground level if connected to the antenna switch relays.

Please note: Output current capability is 500mA per output, but the total dissipation of whole IC must be kept under 1W. The average output Darlington pair saturation voltage is 1.7V, therefore the power supply must deliver at least 13.7V for the 12V relays! More information can be found in the [HTCSEMI HT62783ARTZ Datasheet](#).

The output driver is able to operate from max 50V, but this voltage (over 30V) must be connected externally through JP1 – PIN2. JP1-PIN1 and JP1-PIN3 must be unconnected.

JP1 configuration

PIN1+PIN2 closed	Output drivers are powered from Vcc
PIN2+PIN3 closed	Output drivers are powered from +5V (!!! 5V can provide max 100mA)
PIN1+PIN2+PIN3 opened	connect second power supply for the output drivers through PIN2



Power supply

The ICOM Band Decoder is powered-through J3 header. Serial diode D3 protects the logic circuits against the over polarity. If the input voltage is used for the output drivers too, the voltage drop 0.5V at the diode, has to be taken into the account and you must use higher power supply voltage. 13.8V should be enough for 12V relays. If you want to optionally use 24V relays then use cca 26-27V power supply. Do not USE more than 30V. **+5V stabilizer 78L05 can not accept more than 30V at the input!!**

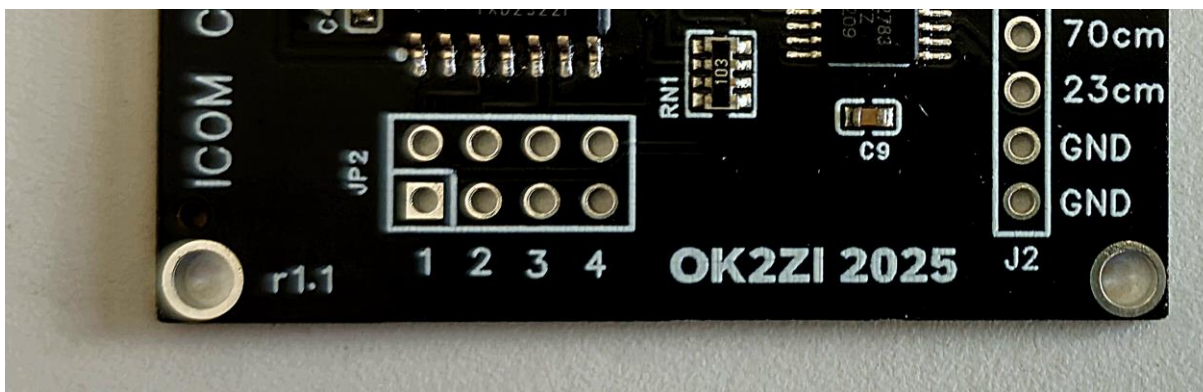
JP1 1+2 = Vcc at the output drivers (USE for the relays switching)

JP1 2+3 = +5V at the output drivers (USE for the BCD output)

Take care about the output voltage selection. Almost all equipment with the BCD inputs accepts only +5V !!!

Decoder configuration

JP2 jumpers are for the decoder configuration.



JP2-4 open default configuration, decoder decodes the band and the band outputs are activated
JP2-4 closed BCD encoding. Decoder reads the frequency and translate it to YAESU BCD code
JP2-1,2,3 communications speed (Baud) selector.

Configured decoder speed must match the CI-V bus speed. Otherwise, the decoder will not work!!

	JP2-1	JP2-2	JP2-3	Bd
0	open	open	open	9600
1	closed	open	open	19200
2	open	closed	open	38400
3	closed	closed	open	2400
4	open	open	closed	115200
5	closed	open	closed	4800
6	open	closed	closed	57800
7	closed	closed	closed	1200

Applications – band decoder, relay driver

With the Band Decoder you can control up to 14 separate antenna relays, one for each band.

Very often many Hams use multiband antennas and then a smaller number of relays are needed. In this case you can combine the Band Decoder outputs in parallel. Because of open emitter driver design, there is no risk of driver damage.

Examples:

1. You have 4:1 antenna switch and 4 antennas, but only 2 are mono-band antennas.

Band	Antenna	4:1 Switch Port	Band Decoder outputs combination
160m	Dipole	1	160m
80m	Inverted Vee	2	80m
40m	40/30 GP	3	40m+30m
30m	40/30 GP	3	40m+30m
20m	5 band Yagi	4	20m+17m+15m+12m+10m
17m	5 band Yagi	4	20m+17m+15m+12m+10m
15m	5 band Yagi	4	20m+17m+15m+12m+10m
12m	5 band Yagi	4	20m+17m+15m+12m+10m
10m	5 band Yagi	4	20m+17m+15m+12m+10m

2. You have 6:1 antenna switch and 6 antennas, and 2 are multi-band antennas.

Band	Antenna	6:1 Switch Port	Band Decoder outputs combination
160m	160m Dipole	1	160m
80m	80m Inverted Vee	2	80m
40m	40m Ground Plane	3	40m
30m	30m Dipole	4	30m
20m	20m-15m-10m Yagi	5	20m+15m+10m
17m	17m-12m Yagi	6	17m+12m
15m	20m-15m-10m Yagi	5	20m+15m+10m
12m	17m-12m Yagi	6	17m+12m
10m	20m-15m-10m Yagi	5	20m+15m+10m

Applications – BCD encoder, BCD coded band output

The ICOM Band Decoder can convert frequency/band info into the BCD code often used by YAESU, Elecraft and other ham radio products. If you have, for example, a set of bandpass filters automatically switched by BCD code from YAESU radio, you can use this decoder to transcode ICOM CI-V frequency code to YAESU BCD band info.

The bands are coded according to this table and common practice defined by YAESU:

	Band	D	C	B	A
0	N/A	0	0	0	0
1	160m	0	0	0	1
2	80m	0	0	1	0
3	40m	0	0	1	1
4	30m	0	1	0	0
5	20m	0	1	0	1
6	17m	0	1	1	0
7	15m	0	1	1	1
8	12m	1	0	0	0
9	10m	1	0	0	1
10	6m	1	0	1	0
11	2m	1	0	1	1
12	70cm	1	1	0	0
13	23cm	1	1	0	1
14	N/A	1	1	1	0
15	60m	1	1	1	1
Outputs activated		30m	40m	80m	160m
		10m	12m	15m	17m
		23cm	70cm	2m	60m

BCD code is mapped to 3x4 outputs according to the table above, so you can connect directly 3 different equipments separately.

Summary specification

Power Supply Voltage	+12 to +30 volts dc
Quiescent Current required (no relays activated)	10 mA
Output current capability	350 mA per output
Total output current capability (for whole Band Decoder)	1000mA (limited by D3)
Dimension	45x50mm

Any suggestion, comments or questions are welcomed at email

OK2ZI@ATLAS.CZ